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UNSW Sydney

TERM 1 2025 FINAL EXAMINATION

ELEC3115: ELECTROMAGNETIC ENGINEERING

1. TIME ALLOWED – 2 hours
2. READING TIME – 10 minutes
3. THIS EXAMINATION PAPER HAS 7 PAGES
4. TOTAL NUMBER OF QUESTIONS – 5
5. TOTAL MARKS AVAILABLE – 100
6. MARKS AVAILABLE FOR EACH QUESTION ARE SHOWN IN THE EXAMINATION PAPER.
 - Assumptions made in answering the questions should be stated explicitly.
 - Candidates must show all their workings to receive any partial marks applicable.
7. ALL ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY BE USED ONLY FOR DRAWING, SKETCHING OR GRAPHICAL WORK
8. ANSWER EACH QUESTION IN A SEPARATE ANSWER BOOK. WRITE YOUR NAME AND QUESTION NUMBER ON THE FRONT PAGE OF EACH ANSWER BOOK. THE STUDENT ID MUST BE FILLED IN CORRECTLY IN EACH EXAM SCAN ANSWER BOOK.
9. THIS PAPER MAY **NOT** BE RETAINED BY CANDIDATE
10. CANDIDATES MAY BRING TO THE EXAMINATION UNSW-Approved Calculator
11. THE FOLLOWING MATERIALS WILL BE PROVIDED: one Smith Chart and one mm-Graph Paper

Question 1 [20 marks]

A two-wire overhead power transmission line has two copper conductors separated by air as the dielectric material, as indicated in Figure Q1. Each conductor has a circular cross-section with a radius of $a = 2.5$ mm, and the separation between the conductors is $d = 0.65$ m.

- (i) [7 marks] Using Gauss's law, derive an expression for the capacitance between the two conductors and, hence, calculate the capacitance of the cable pair per meter.
- (ii) [3 marks] Considering the air between the two conductors as a lossy dielectric, calculate the leakage resistance of the conductor per meter. The conductivity of air is 1.4×10^{-15} S/m, and the relative permittivity is 1.
- (iii) [7 marks] Using Gauss's law, derive an expression for the maximum E field in terms of the voltage between the conductors and hence calculate the maximum voltage the cable can be operated at, assuming that the maximum allowable field within the dielectric is not to exceed 50% of its breakdown value. Assume that the air dielectric between the conductors has a breakdown strength (E_{max}) of 3 MV/m.
- (iv) [3 marks] What are the limitations or possible errors associated with applying Gauss's law to calculate capacitance for the given case of Figure Q1? Explain why the method of images could be used to find the electric field between the conductors, and hence their capacitance, with higher accuracy.

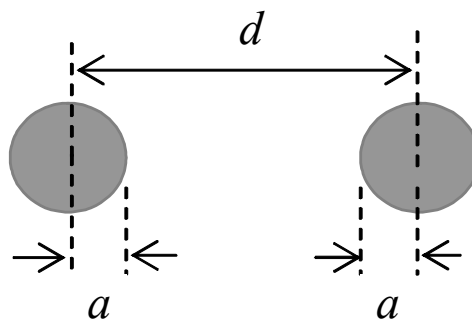
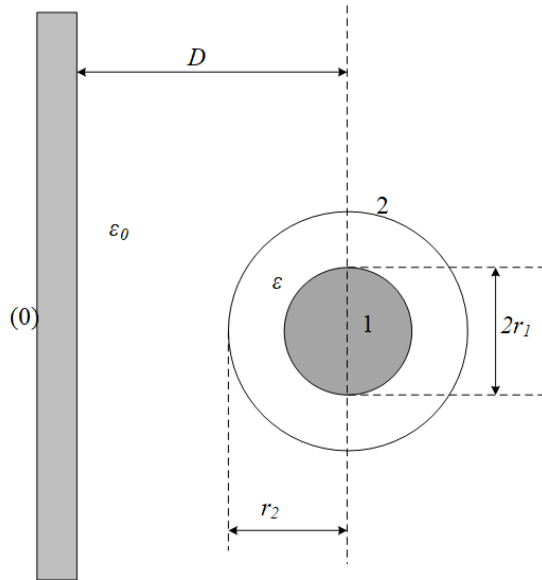


Figure Q1

See next page for question 2

Question 2 [20 marks]

A cross-section of a system comprising a communication coaxial cable of length l , running parallel to a conducting wall at zero potential, is shown in Figure Q2.



Length of the cable $l = 8$ m,
Distance from the centre of the cable to the
conducting wall $D = 2$ cm,
Outer radius of the cable $r_2 = 5$ mm,
Inner radius of the cable $r_1 = 1$ mm and
permittivity of the cable dielectric $\epsilon = 3\epsilon_0$ F/m.

Figure Q2

- (a) [6 marks] Find the values of the capacitance between conductor 1 and shield 2 and between shield 2 and the wall (represented as (0) in Figure Q2).
- (b) [2 marks] The metallic shield (shown as 2 in Figure Q2) of the cable, at radius r_2 , is connected to the ground for electrostatic shielding. Explain how electrostatic shielding prevents capacitive coupling between conductor 1 and the wall.
- (c) [7 marks] Using Laplace's equation, find an expression for the electric field in terms of the voltage V_0 between conductor 1 and shield 2. The shield is grounded and at zero potential.
- (d) [5 marks] If 15V is applied between the conductor 1 and shield 2, and the insulating jacket is made of rubber with a breakdown strength of 400 kV/m and a permittivity of $\epsilon = 3\epsilon_0$ F/m, do the rubber insulation and the air outside the cable break down? The breakdown strength of air is 3MV/m, and its permittivity is ϵ_0 . Explain using a plot of the electric field versus radial distance from the centre of the cable. You may consider using the boundary conditions of the electric field at the interface of two dielectrics.

See next page for question 3

Question 3 [20 marks]

A permanent magnet specimen is magnetised by a strong electromagnet created by a coil, as indicated in Figure Q3 (a). The three keeper sections, A, B, and C, hold the magnet in place, and they are infinitely permeable. The dimensions of the magnet and the keeper sections are given below: $l_m = 8 \text{ mm}$, $d = 30 \text{ mm}$, $l_c = 25 \text{ mm}$, $a = 20 \text{ mm}$, $b = 40 \text{ mm}$, $c = 40 \text{ mm}$

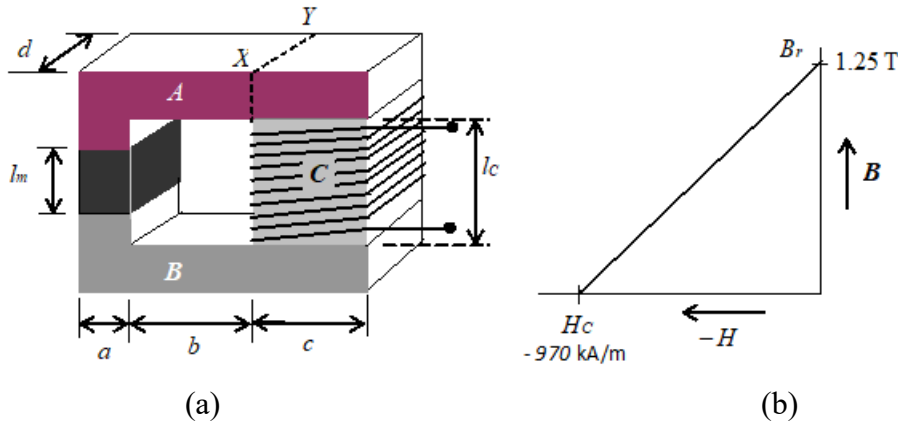


Figure Q3

The cross-sections of the magnet and the keepers to the left of the section XY are uniform. The depths of the magnet specimen and the keepers are the same, as shown in Figure Q3(a) as d .

- (i) [4 marks] To magnetise the piece of the magnet, the coil needs to produce a flux density B equal to 1.28 T on the magnet piece. If the coil has 1000 turns, how much current needs to flow through the coil? The permeability of the magnet is the same as the air. Use the magnetic circuit method for this calculation.
- (ii) [3 marks] After full magnetisation, the coil current is reduced to zero, and the magnet is then held *in place* by the three keeper sections, A, B, and C, without the coil. Find the magnetic flux density B and the intensity H in the magnet using the B-H curve of the magnet given in Figure Q3 (b).
- (iii) [7 marks] The Keeper section C is removed subsequently, as a first step to removing the magnet for installing in an equipment. The keeper C is essentially air after the infinitely permeable metal piece is removed. Draw the load line for the air gap on the BH curve of the magnet and, using the graphical method, find the B and H fields in the magnet. Use the mm graph paper provided for accurate estimation
- (iv) [6 marks] Find the stored energies in the magnet and the air gap. Where do these energies come from?

See next page for question 4

Question 4 [25 marks]

Consider a lossless coaxial cable of length $l = 0.5$ m, filled by a non-magnetic dielectric.

The cable possesses an inductance per unit length $\mathcal{L} = 1$ $\mu\text{H}/\text{m}$ and a capacitance per unit length $\mathcal{C} = 100$ pF/m.

The cable is terminated by a resistor of value $R_L = 20$ Ω , and fed by an DC voltage generator with internal resistance $R_G = 20$ Ω and voltage $v_G = 1$ V. An ideal switch is interposed between the generator and the cable, as shown in Figure Q4 below.

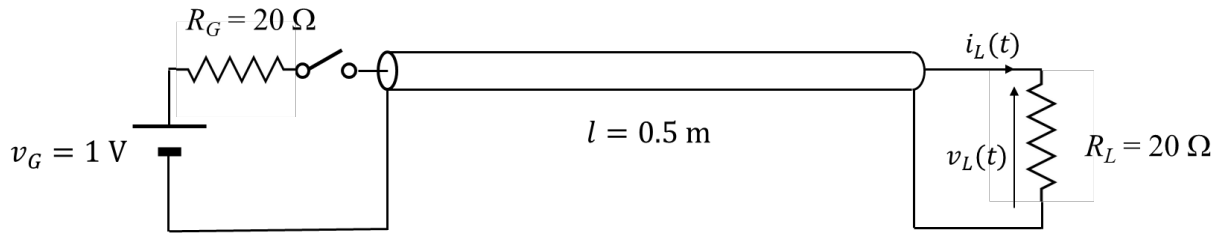


Figure Q4

At time $t = 0$, the switch is closed and the generator is connected to the cable. For the time interval $t = 0 - 30$ ns draw:

- (i) **[12 marks]** The reflection diagram, which must contain the numerical values of the voltage and current on every path, and a correct labelling of the time and distance axes.

[Note: even though we do not ask to calculate the current at the load in part (ii) below, the reflection diagram in part (i) above must also include the values of the currents on each path]

- (ii) **[6 marks]** The voltage at the load, $v_L(t)$.

- (iii) **[3 marks]** Construct and draw a lumped-element circuit that best approximates the behaviour of the “real” circuit. Describe the reasoning you use to construct this circuit and provide the numerical value of its parameters.

- (iv) **[4 marks]** Draw the voltage at load $v_L(t)$ in the time interval $t = 0 - 30$ ns in the lumped-elements circuit drawn at point (iii) above. Discuss the difference or similarity between the “true” voltage drawn at point (ii), and the one you obtain here.

See next page for question 5

Question 5 [15 marks]

Consider a coaxial cable, filled with a non-magnetic dielectric with $\epsilon_r = 4$. With this dielectric, the cable has a characteristic impedance $Z_0 = 50 \Omega$. The cable is 1 m long, terminated on the right by an impedance $Z_L = 20 + j100 \Omega$, and connected on the left to a voltage generator with internal resistance $R_G = 50 \Omega$ that produces an AC voltage of amplitude $V_G = 1 \text{ V}$ and frequency $f = 400 \text{ MHz}$.

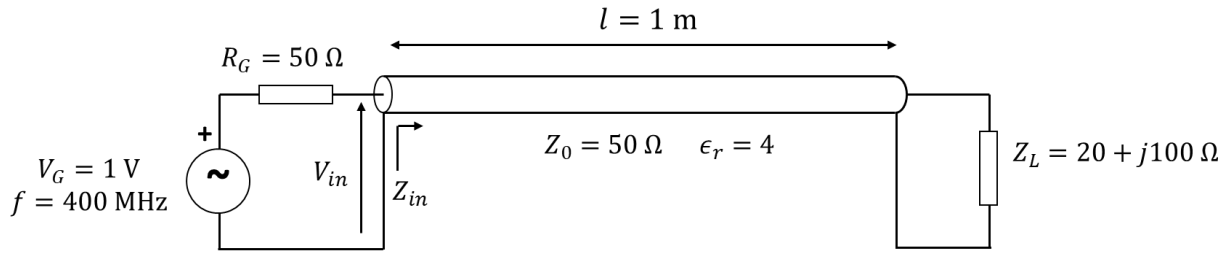


Figure Q5

- (i) [3 marks] Calculate the wavelength λ of the propagating signal in the cable.
- (ii) [2 marks] Calculate the voltage standing wave ratio VSWR in the cable.
- (iii) [8 marks] Using the supplied Smith chart, calculate the input impedance Z_{in} seen by the generator at the entrance of the cable (to the right of its own internal resistance). Provide a clear written explanation of the steps you take while using the Smith chart to arrive at the result, and attach the completed Smith chart to your exam paper.
- (iv) [2 marks] Calculate the voltage V_{in} at the input of the cable.

See next page for useful formula of Part A

Useful formula and constants for part A:

$\epsilon_0 = 8.854 \times 10^{-12} \text{ [F/m]}$ $\vec{\nabla} \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$ $\vec{\nabla} \times \vec{A} = \vec{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \vec{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \vec{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$ $\oint_s \vec{E} \cdot d\vec{s} = \frac{Q_{enclosed}}{\epsilon} \quad \text{or} \quad \oint_s \vec{D} \cdot d\vec{s} = Q_{enclosed}$ $Q_{enclosed} = \int_v \rho dv \quad \text{or} \quad \int_s \rho_s ds \quad \text{or} \quad \int_l \rho_l dl$ $V_{ba} = - \int_a^b \vec{E} \cdot d\vec{l}$ $\vec{P} = \vec{D} - \epsilon_0 \vec{E}, \quad \vec{D} = \epsilon_0 \epsilon_r \vec{E}$ $\vec{E} = \frac{\rho_l}{2\pi\epsilon_0 r} \vec{a}_r$ $V_{2l} = \frac{\rho_l}{2\pi\epsilon_0} \ln \left(\frac{r_l}{r_2} \right)$ $ E_{lt} = E_{2t} , \quad D_{ln} = D_{2n} $ $C = \frac{\rho_l}{V_{l2}} = \frac{\pi\epsilon_0}{\ln \frac{d-a}{a}} \text{ [F/m]}, \quad \text{If } d \gg a, \quad C \approx \frac{\pi\epsilon_0}{\ln \frac{d}{a}} \text{ [F/m]}$ $C = \frac{\rho_l}{V_{ab}} = \frac{2\pi\epsilon}{\ln \left(\frac{b}{a} \right)} \text{ [F/m]}$ $C = \frac{\pi\epsilon}{\ln \left[\left(\frac{d}{2a} \right) + \sqrt{\left(\frac{d}{2a} \right)^2 - 1} \right]} \text{ F/m}$ $w_e = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \frac{D^2}{\epsilon} \text{ [J/m}^3\text{]}$ $\vec{F}_e = \pm \vec{\nabla} W_e$ $\vec{\nabla}^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \theta^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho}{\epsilon}$	$\vec{J} = \rho \vec{u}, \quad I = \int_s \vec{J} \cdot d\vec{s} \quad \text{A}$ $\vec{J} = \sigma \vec{E}, \quad R = \frac{l}{\sigma S}, \quad P = (\vec{E} \cdot \vec{J}) \int_{vol} dv$ $\vec{\nabla} \cdot \vec{J} = -\frac{\partial \rho}{\partial t}, \quad \rho = \rho_0 e^{-\left(\frac{\sigma}{\epsilon}\right)t}$ $\vec{\nabla} \times \vec{J} = 0$ $RC = \frac{\epsilon}{\sigma}$
	$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]}$ $\oint_c \vec{B} \cdot d\vec{l} = \mu_0 I_c, \quad \oint_c \vec{H} d\vec{l} = I_{enclosed}$ $\vec{B} = \vec{\nabla} \times \vec{A},$ $\vec{B} = \mu_0 \vec{H} + \mu_0 \vec{M},$ $\vec{B} = \mu_0 (1 + \chi_m) \vec{H} = \mu_0 \mu_r \vec{H} = \mu \vec{H}$ $\vec{\nabla}^2 \vec{A} = -\mu_0 \vec{J}$ $\Phi = \int_s \vec{B} \cdot d\vec{s}, \quad \Re = \frac{l}{\mu S},$ $L = \frac{N\phi}{i}, \quad L_{l2} = \frac{N_2 \phi_{l2}}{I_l}$ $W_m = \frac{1}{2} \int_{vol} \vec{H} \cdot \vec{B} dv, \quad \vec{F} = \pm \vec{\nabla} W_m,$
	$EMF = -N \frac{d\phi}{dt}, \quad EMF = \oint_c (\vec{u} \times \vec{B}) \cdot d\vec{l}$ $\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$ $V_{rms} = 4.44 f N B_m S_c$ $\delta = \sqrt{\frac{2}{\sigma \mu \omega}}, \quad R_{ac} = \frac{l}{\delta \sigma} \left(\frac{l}{\pi(D - \delta)} \right)$ $P_h = k_2 B_m^n f, \quad P_e = k_1 B_m^2 f^2$

END OF EXAM PAPER